

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-I]

TEST # 15

DATE : 29-12-2019

REVISION TEST

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

Corporate Office

ALLEN CAREER INSTITUTE

"SANKALP", CP-6, Indra Vihar, Kota (Rajasthan)-324005

☎ +91-744-2757575 ☎ +91-744-2435003 ✉ info@allen.ac.in

Rajkot Centres

Kalawad Road

"ISHAVASYAM", B/h. Crystal Mall, Opp./O.C. Quarters,
Kalawad Road, Rajkot (Gujarat)-360005

☎ 0281-2562131 ☎ +91-85111-43783 ✉ rajkot@allen.ac.in

Limbudiwadi Centre

Opp. Landmark 2 Apartment, Limbudiwadi
Main Road, Off. Kalawad Road, Rajkot (Gujarat)- 360007

☎ 0281-2450228

www.allen.ac.in

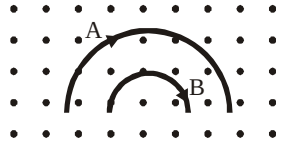
Your Target is to secure Good Rank in NEET 2020

SPACE FOR ROUGH WORK

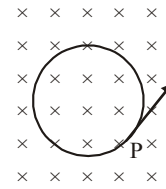
HAVE CONTROL ü HAVE PATIENCE ü HAVE CONFIDENCE Ğ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

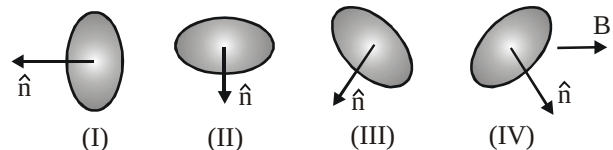
1. Two particles A and B of masses m_A and m_B respectively and having the same charge are moving in a plane. A uniform magnetic field exists perpendicular to this plane. The speeds of the particles are v_A and v_B respectively, and the trajectories are as shown in the figure. Then



- (1) $m_A v_A < m_B v_B$
 (2) $m_A v_A > m_B v_B$
 (3) $m_A < m_B$ and $v_A < v_B$
 (4) $m_A = m_B$ and $v_A = v_B$
2. Time period of thin bar magnet of vibration magnetometer is 'T'. If it cuts into two equal parts in then the time period of each part follows.
- (a) Along or parallel to its length
 (b) Perpendicular to its length,
- (1) T, T (2) T, T/2
 (3) T/2, T (4) T/2, T/2
3. A particle having a charge of $10.0 \mu\text{C}$ and mass $1 \mu\text{g}$ moves in circle of radius 10 cm under the influence of a magnetic field of induction 0.1 T. When the particle is at a point P, a uniform electric field is switched on so that the particle starts moving along the tangent with a uniform velocity. The electric field is



- (1) 0.1 V/m (2) 100 V/m
 (3) 10.0 V/m (4) 1.0 V/m
4. Following figure shows the path of an electron that passes through two regions containing uniform magnetic field of magnitudes B_1 and B_2 . Its path in each region is a half circle, choose the correct option :-
-
- (1) B_1 is into the page and it is stronger than B_2
 (2) B_1 is into the page and it is weaker than B_2
 (3) B_1 is out of the page and it is weaker than B_2
 (4) B_1 is out of the page and it is stronger than B_2
5. The figure represents four positions of a current carrying coil in a magnetic field directed towards right. $\hat{n}$ represents the direction of area vector of the coil. The correct order of potential energy is:-



SPACE FOR ROUGH WORK

- (1) $I > III > II > IV$
 (2) $I < III < II < IV$
 (3) $IV < I < II < III$
 (4) $III > II > IV > I$
6. A horizontal rod of mass 10 gm and length 10 cm is placed on a smooth plane inclined at an angle of 60° with the horizontal, with the length of the rod parallel to the edge of the inclined plane. A uniform magnetic field of induction B is applied vertically downwards. If the current through the rod is 1.73 A, then the value of B for which the rod remains stationary on the inclined plane is
- (1) 1.73 T (2) $\frac{1}{1.73}$ T
 (3) 1T (4) None of these
7. At any place on surface of earth angle of dip in two vertical planes at right angle to each other is 30° and 45° respectively, then true angle of dip at that place will be :-
- (1) $\cot^{-1}\left(\frac{1}{2}\right)$ (2) $\tan^{-1}\left(\frac{1}{2}\right)$
 (3) $\tan^{-1}(2)$ (4) $\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$
8. Electron moves right angles to magnetic field of 4.4×10^{-2} T with a speed of 6×10^7 m/s. If the specific charge of the electron is 1.7×10^{11} C/kg, then number of cycles in one second will be.
- (1) 119×10^6 (2) 6.28×10^6
 (3) 1.19×10^8 (4) 1.19×10^9
9. An electron is moving in a circular orbit in a magnetic field of 1.0×10^{-4} Wb/m². Its period of revolution is :-
- (1) 3.5×10^{-7} sec (2) 7.0×10^{-7} sec
 (3) 1.05×10^{-6} sec (4) 2.1×10^{-6} sec
10. A long straight wire of radius R carries a steady current I . The current is uniformly distributed over its cross-section. Magnetic field at distance $2R$ from its axis is B_1 and at distance $R/3$ inside the wire from its surface is B_2 . The ratio B_1/B_2 will be
- (1) $\frac{3}{2}$ (2) $\frac{2}{3}$ (3) $\frac{3}{4}$ (4) $\frac{4}{3}$
11. A long straight wire of radius 'a' carries a steady current I . The current is uniformly distributed over its cross-section. Magnetic field at distance $2a$ from axis is B . Then another distance from axis at which magnetic field is equal to 'B' will be :-
- (1) a (2) $\frac{a}{4}$ (3) $\frac{a}{2}$ (4) $4a$
12. The relative permeability is represented by μ_r and susceptibility is denoted by χ for a magnetic substance. For a paramagnetic substance :-
- (1) $\mu_r < 1, \chi < 0$ (2) $\mu_r < 1, \chi > 0$
 (3) $\mu_r > 1, \chi < 0$ (4) $\mu_r > 1, \chi > 0$
13. An equilateral triangle is made by a wire of length ' ℓ '. It is hanging between two poles of permanent magnets such that magnetic field $\vec{B}$ acting in plane of loop. If current i is flowing in the loop and torque τ act on the loop, then value of ' ℓ ' will be :-

SPACE FOR ROUGH WORK

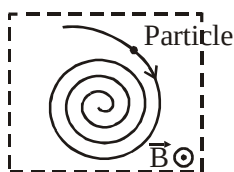
(1) $\frac{2}{\sqrt{3}} \left(\frac{\tau}{Bi} \right)$ (2) $\frac{2}{\sqrt{3}} \left(\frac{\sqrt{3}\tau}{Bi} \right)^{1/2}$

(3) $2 \left(\frac{3\sqrt{3}\tau}{Bi} \right)^{1/2}$ (4) $\frac{2}{\sqrt{3}} \left(\frac{\tau}{Bi} \right)^{1/2}$

14. In absence of external magnetic field, Magnetic moment of diamagnetic, paramagnetic and ferromagnetic materials is M_d , M_p , M_f , then

- (1) $M_d = 0$ and $M_p \neq 0$
(2) $M_p \neq 0$ and $M_f \neq 0$
(3) $M_d \neq 0$ and $M_f \neq 0$
(4) $M_p = 0$ and $M_f = 0$

15. A uniform magnetic field is directed out of the page. A charge particle, moving in the plane of page, following a clockwise spiral of decreasing radius as shown. A responsible explanation is.



- (1) The charge is positive and slowing down
(2) The charge is negative and slowing down
(3) The charge is positive and speeding up
(4) The charge negative and speeding up
16. In a mass spectrometer used for measuring the mass of ions, the ions are initially accelerated by an electric potential V and then made to describe semicircular paths of radius R using a

magnetic field B . If V and B are kept constant, the ratio $\left(\frac{\text{mass of the ion}}{\text{charge on the ion}} \right)$ will be proportional to :

- (1) R (2) $\frac{1}{R}$ (3) $\frac{1}{R^2}$ (4) R^2

17. For which magnetic material magnetic susceptibility is $\chi \propto \frac{1}{T}$ (T-absolute temperature)

- (1) Diamagnetic (2) Paramagnetic
(3) Ferromagnetic (4) Non magnetic

18. Cobalt shows ferromagnetic property at room temperature. If the temperature is less than Curie temperature then it will show :

- (1) diamagnetism
(2) paramagnetism
(3) anti ferromagnetism
(4) ferromagnetism

19. We heat a magnetic material above Curie temperature, there is no change in magnetic properties, then magnetic material will be :

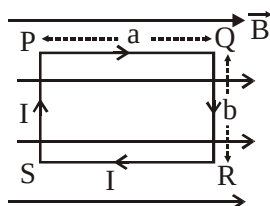
- (1) paramagnetic (2) diamagnetic
(3) ferromagnetic (4) none of these

20. A permanent magnet is made by steel, because steel has

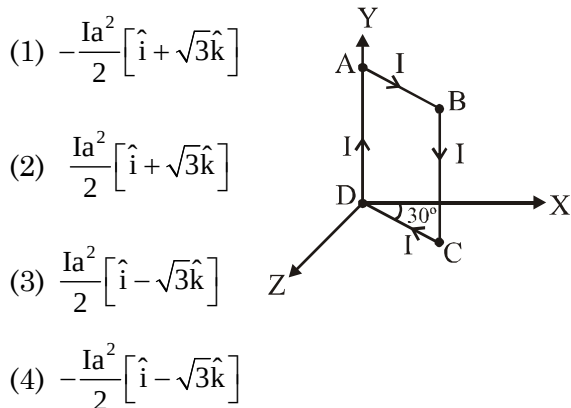
- (1) High retentivity and low coercivity
(2) Low retentivity and high coercivity
(3) High retentivity and high coercivity
(4) Low retentivity and low coercivity

SPACE FOR ROUGH WORK

21. A rectangular loop PQRS in which current I flowing is placed in uniform magnetic field $\vec{B}$ as shown in figure. Incorrect statement will be:-

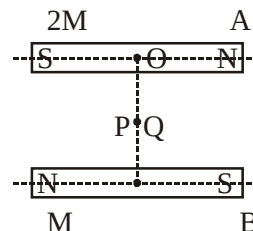


- (1) Force on side QR is IbB , perpendicular outward to the plane of paper.
 - (2) Force on side PQ is IaB , perpendicular outward to the plane of paper.
 - (3) Net force on loop is zero.
 - (4) Force on side RS will be zero.
22. A square loop of side 'a' carries a current I . It is placed as shown in figure. Magnetic moment of the loop will be :-



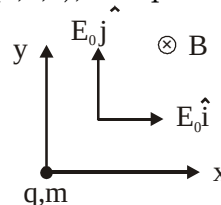
- (1) $-\frac{Ia^2}{2}[\hat{i} + \sqrt{3}\hat{k}]$
 - (2) $\frac{Ia^2}{2}[\hat{i} + \sqrt{3}\hat{k}]$
 - (3) $\frac{Ia^2}{2}[\hat{i} - \sqrt{3}\hat{k}]$
 - (4) $-\frac{Ia^2}{2}[\hat{i} - \sqrt{3}\hat{k}]$
23. Which of the following is incorrect for diamagnetic material :-
- (1) $\chi_m < 0$
 - (2) $\mu < \mu_0$
 - (3) $0 < \mu_r < 1$
 - (4) χ_m depends on temperature.

24. Two bar magnet (A and B) of magnetic moment $2M$ and M , are placed parallel to each other as shown in figure, distance between their centres is 'd'. A charge Q is placed at mid point 'P', then force on charge Q will be



- (1) along to the direction of magnetic moment of bar magnet A
 - (2) along to the direction of magnetic moment of bar magnet B
 - (3) along PO
 - (4) will be zero.
25. A free particle of charge q and mass m is at rest at the origin as shown in the figure.

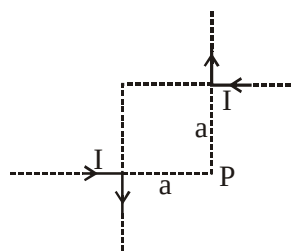
A constant electric field $E_0\hat{i} + E_0\hat{j}$ and constant magnetic field $B(-\hat{k})$ is present in whole region. When the particle reaches at position $(2,2,0)$, its speed will be:-



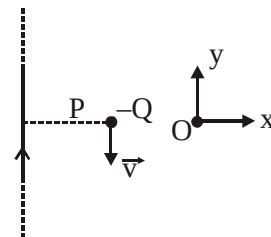
- (1) $\sqrt{\frac{8qE_0}{m}}$
- (2) $\sqrt{\frac{4qE_0}{m}}$
- (3) $\sqrt{\frac{qE_0}{m}}$
- (4) $\sqrt{\frac{2qE_0}{m}}$

SPACE FOR ROUGH WORK

26. Magnetic field at point 'P' due to given current distribution :-

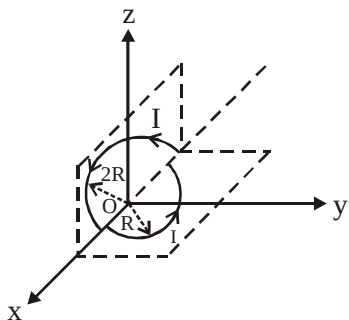


- (1) $\frac{\mu_0 I}{4\pi a} \odot$ (2) $\frac{\mu_0 I}{2\pi a} \odot$
(3) $\frac{\mu_0 I}{\pi a} \otimes$ (4) Zero
27. The work done in rotating a bar magnet of magnetic moment M from its unstable equilibrium position to its stable equilibrium position in a uniform magnetic field ' B ' is :-
(1) $2MB$ (2) MB
(3) $-2MB$ (4) $-MB$
28. Curie temperature is the temperature above which-
(1) a ferromagnetic material becomes paramagnetic
(2) a paramagnetic material becomes diamagnetic
(3) a ferromagnetic material becomes diamagnetic
(4) a paramagnetic material becomes ferromagnetic
29. In given figure current I is flowing in a long straight wire. At the instant when charge $-Q$ on point P starts motion with velocity $\vec{v}$, force acting on it will be.



- (1) In OX direction
(2) In OY direction
(3) Opposite to the OX direction
(4) Opposite to the direction of OY
30. A square loop of side ' a ' carrying current ' I ' is placed in uniform magnetic field ' B ' such that it is in stable equilibrium. Work done to rotate it about an axis perpendicular to its plane and passing through the centre by an angle 180° will be:-
-
- (1) Ia^2B (2) $2Ia^2B$
(3) zero (4) None of these
31. A current loop consists of two semi-circular parts of radius R and $2R$. The semi-circular part of radius R is placed in the XY plane and the other part of radius $2R$ is placed in the XZ plane. If current in the loop is I . The resultant magnetic field due to the two semi-circular parts at their common centre is :-

SPACE FOR ROUGH WORK



- (1) $\frac{\mu_0 I}{2R}$ (2) $\left(\frac{\sqrt{3}\mu_0 I}{2R}\right)$
 (3) $\frac{\sqrt{3}\mu_0 I}{4R}$ (4) $\frac{\sqrt{5}\mu_0 I}{8R}$

32. A short bar magnet of magnetic moment 0.6 J T^{-1} is placed in a uniform magnetic field of 0.32 T . Magnet is in unstable equilibrium when the potential energy is :

- (1) 0.192 J (2) -0.192 J
 (3) zero (4) 0.096 J

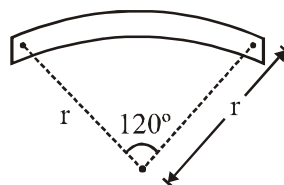
33. An alternating electric field, of frequency 50 Hz is applied across the dees (radius- R) of a cyclotron that is being used to accelerate proton (mass $=m$). The operating magnetic field ' B ' used in the cyclotron and the maximum speed of proton beam, produced by it, are respectively:

- (1) $B = \frac{100\pi m}{e}$ and $v = 100\pi R$
 (2) $B = \frac{50\pi m}{e}$ and $v = 200\pi R$

(3) $B = \frac{50\pi m}{e}$ and $v = 100\pi R$

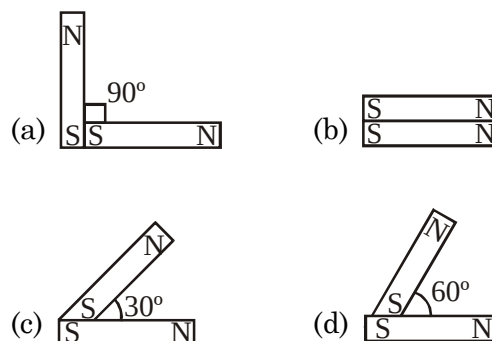
(4) $B = \frac{100\pi m}{e}$ and $v = 200\pi R$

34. A bar magnet of length ' l ' and magnetic dipole moment ' M ' is bent in the form of an arc as shown in figure. The new magnetic dipole moment will be:



- (1) $\frac{\sqrt{3}}{2\pi} M$ (2) $\frac{3\sqrt{3}}{4\pi} M$
 (3) $\frac{3}{4\pi} M$ (4) $\frac{3\sqrt{3}}{2\pi} M$

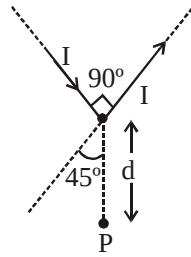
35. Following figures shown the arrangement of bar magnets in different configurations. Each magnet has magnetic dipole moment $\vec{M}$. Which configuration has minimum net magnetic dipole moment ?



- (1) In (a) (2) In (b) (3) In (c) (4) In (d)

SPACE FOR ROUGH WORK

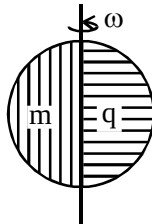
36. Magnetic field at point 'P' due to the arrangement shown of two current carrying long wires.



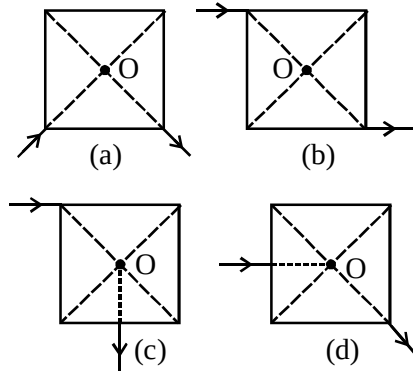
- (1) $\frac{\mu_0 I}{\sqrt{2}\pi d} \left(1 - \frac{1}{\sqrt{2}}\right) \otimes$ (2) $\frac{2\mu_0 I}{\sqrt{2}\pi d} \otimes$
(3) $\frac{\mu_0 I}{\sqrt{2}\pi d} \otimes$ (4) $\frac{\mu_0 I}{\sqrt{2}\pi d} \left(1 + \frac{1}{\sqrt{2}}\right) \otimes$

37. Consider a hypothetic spherical body. The body is cut into two parts about the diameter. One of hemispherical portion has mass distribution 'm' while the other portion has identical charge distribution 'q'. The body is rotated about the axis with constant speed ' ω '. Then the ratio of magnetic moment to angular momentum is :-

- (1) $\frac{q}{2m}$
(2) $> \frac{q}{2m}$
(3) $< \frac{q}{2m}$
(4) cannot be calculated



38. In which of the following square current carrying loop, magnetic field at its centre in non-zero.



- (1) In 'a' (2) In 'b' (3) In 'c' (4) In 'd'
39. Two similar coils of radius R are lying concentrically with their planes at right angles to each other. The current flowing in them are 2I and 3I respectively. The resultant magnetic field induction at the centre will be :

- (1) $\frac{\mu_0 I}{2R}$ (2) $\frac{5\mu_0 I}{2R}$
(3) $\frac{\sqrt{7}\mu_0 I}{2R}$ (4) $\frac{\sqrt{13}\mu_0 I}{2R}$

40. An α -particle (mass = m) is accelerated by using cyclotron. Radius of dees of it is R. α -particle revolving in dees with frequency n, then maximum kinetic energy of the α -particle beam will be :
(1) $\pi m^2 n R^2$
(2) $2\pi^2 n^2 m R^2$
(3) $2\pi^2 n m R^2$
(4) $\pi^2 n^2 m R^2$

SPACE FOR ROUGH WORK

41. The magnetic force acting on a charged particle having charge $-3\mu\text{C}$ in a magnetic field of 2T acting in $-Y$ direction, when the particle velocity is $(3\hat{j} + \hat{k}) \times 10^6 \text{ m s}^{-1}$

- (1) 6N in X -direction
- (2) 6N $-X$ direction
- (3) 18N in X -direction
- (4) 18N in $-Y$ direction

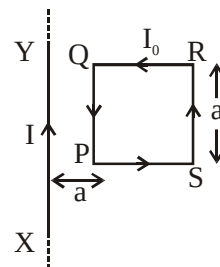
42. Ions of mass m and charge q are accelerated by electric potential V . Now they enter in a uniform magnetic field ' B ' perpendicular to it and ions moves in circular path. Radius of circular path will be :

- (1) $\frac{mV}{qB^2}$
- (2) $\frac{1}{B} \sqrt{\frac{2mV}{q}}$
- (3) $\sqrt{\frac{2mV}{qB}}$
- (4) $2\sqrt{\frac{mV}{qB^2}}$

43. A circular loop is made by a wire of length ' ℓ '. It is placed in a uniform magnetic field ' B '. When current is flows in it, maximum torque τ is act on it, then value of following current in it.

- (1) $\frac{2\pi^2\tau}{\ell^2B}$
- (2) $\frac{\pi\tau}{2\ell^2B}$
- (3) $\frac{4\pi\tau}{\ell^2B}$
- (4) $\frac{2\pi\tau}{\ell^2B}$

44. A square loop PQRS carrying current I_0 , is placed near and coplanar with a long straight conductor XY carrying current I . The net force on the loop will be.

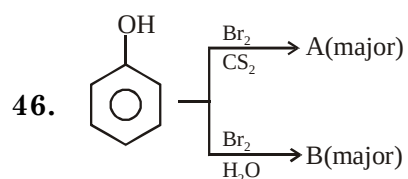


- (1) $\frac{\mu_0 I_0 I}{2\pi}$ attractive
- (2) $\frac{\mu_0 I_0 I}{4\pi}$ repulsive
- (3) $\frac{\mu_0 I I_0}{2\pi}$ repulsive
- (4) $\frac{\mu_0 I I_0}{4\pi}$ attractive

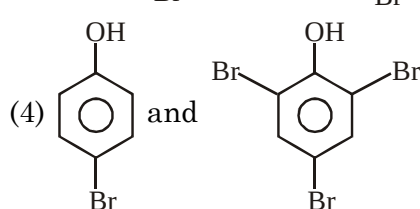
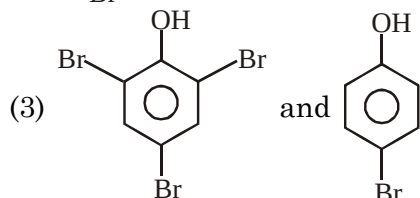
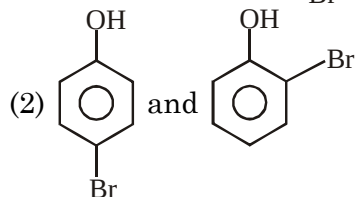
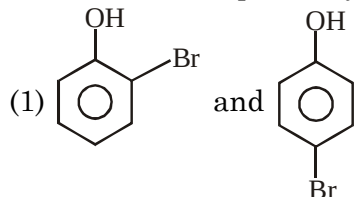
45. The angle of dip on a given meridian, making an angle 30° with the magnetic meridian is equal to 30° . The true angle of dip at the given point is :-

- (1) $\tan^{-1}\left(\frac{1}{2}\right)$
- (2) $\tan^{-1}(2)$
- (3) $\tan^{-1}(\sqrt{2})$
- (4) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$

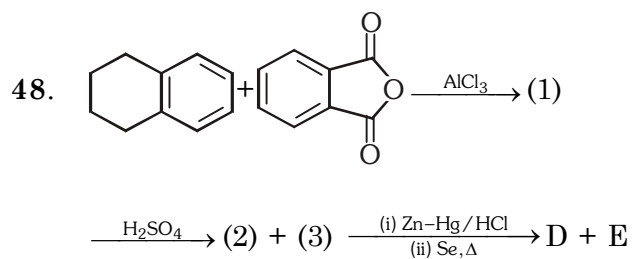
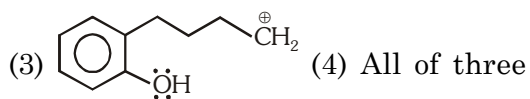
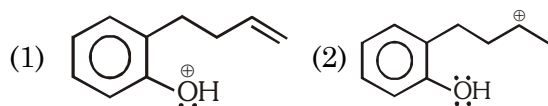
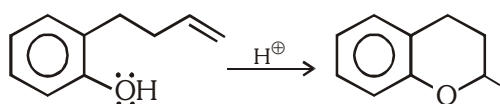
SPACE FOR ROUGH WORK



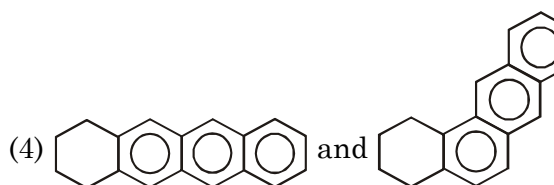
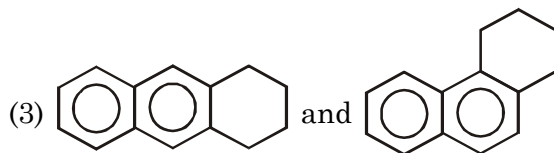
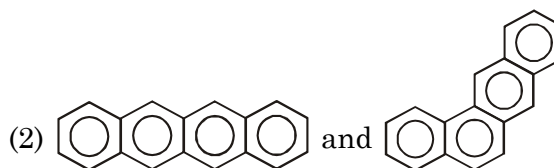
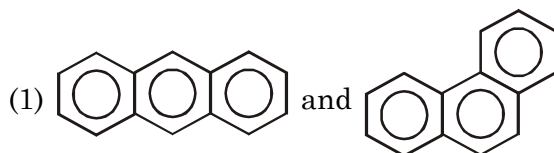
A and B are respectively :



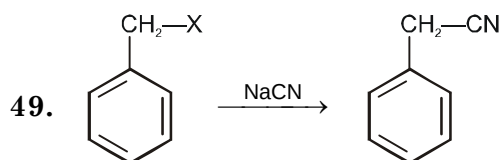
47. Which of the following ion is formed in the following reaction :



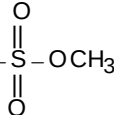
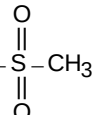
D and E are :



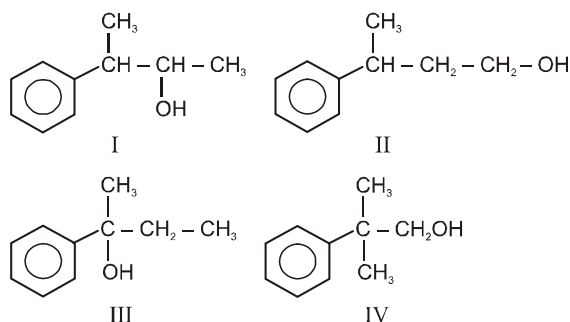
SPACE FOR ROUGH WORK



In the given reaction rate is fastest, when (X) is :

- (1) -OH (2) -NH₂
(3)  (4) 

50. The relative rate of acid catalysed dehydration of following alcohols would be,



- (1) III > I > IV > II (2) III > IV > I > II
(3) I > III > IV > II (4) III > IV > I > II

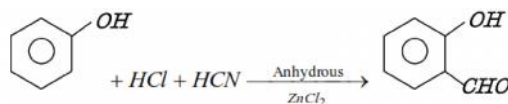
51. Benzyl alcohol is obtained from benzaldehyde by

- (1) Fittig's reaction
(2) Cannizaro's reaction
(3) Kolbe's reaction
(4) Wurtz's reaction

52. Benzene diazonium chloride on boiling with dilute sulphuric acid gives

- (1) Toluene (2) Benzoic acid
(3) Benzene (4) Phenol

53. The following reaction :



is known as

- (1) Perkin reaction
(2) Gattermann reaction
(3) Kolbe reaction
(4) Aldol reaction

54. The compound obtained by heating salicylic acid with phenol in the presence of phosphorus oxychloride is

- (1) Salol
(2) Aspirin
(3) Oil of wintergreen
(4) o-chlorobenzoyl chloride

55. Phenol $\xrightarrow[\text{H}^+]{\text{CHCl}_3/\text{NaOH}}$ Salicylaldehyde

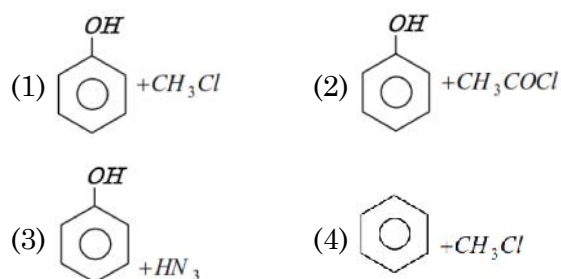
The above reaction is known as

- (1) Reimer Tiemann reaction
(2) Bucherer reaction
(3) Gattermann synthesis
(4) Perkin reaction

56. When Benzene diazonium salt reacts with H₃PO₂ it gives

- (1) Benzene
(2) Chloro benzene
(3) Both (1) and (2)
(4) None of these

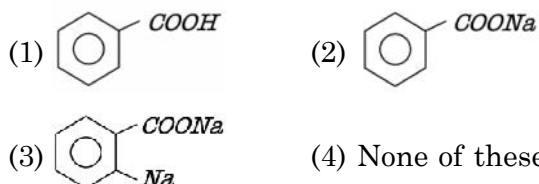
57. In Friedal-Crafts acylation, besides AlCl₃, the other reactants are



SPACE FOR ROUGH WORK

58. When phenyl magnesium bromide reacts with *t*-butanol, the product would be
(1) Benzene (2) Phenol
(3) *t*-butyl benzene (4) *t*-butyl ether

59. Toluene reacts with excess of Cl_2 in presence of sunlight to give a product which on hydrolysis followed by reaction with NaOH gives



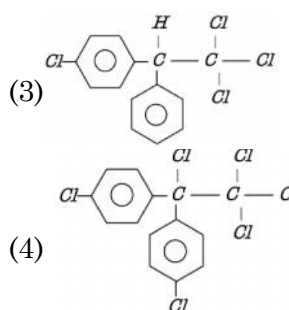
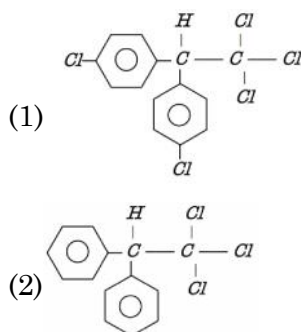
60. Reaction of alkyl halides with aromatic compounds in presence of anhydrous AlCl_3 is known as

- (1) Friedal-Craft reaction
(2) Hofmann degradation
(3) Kolbe's synthesis
(4) Beckmann rearrangement

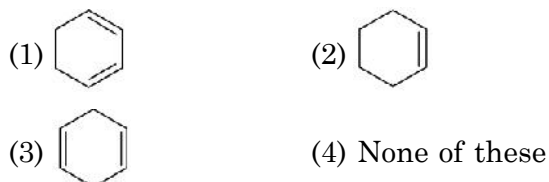
61. When benzene is heated with chlorine in the presence of sunlight, it forms

- (1) Benzene Hexa Chloride
(2) Cyclopropane
(3) *p*-dichlorobenzene
(4) None of these

62. Which one of the following is the correct formula of dichlorodiphenyl trichloroethane



63. 1, 2 di-bromo cyclohexane on dehydro halogenation gives



64. As compared to potassium, sodium has

- (1) Lower electronegativity
(2) Higher ionization potential
(3) Greater atomic radius
(4) Lower melting point

65. Based on lattice energy and other considerations which one of the following alkali metal chlorides is expected to have the highest melting point

- (1) CsCl (2) NaCl
(3) KCl (4) RbCl

66. The reagent commonly used to determine hardness of water is

- (1) Oxalic acid
(2) Disodium salt of *EDTA*
(3) Sodium citrate
(4) Sodium thiosulphate

67. K, Ca metals Li may be arranged in the decreasing order of their standard electrode potentials as

- (1) *K, Ca, Li* (2) *Li, K, Ca*
(3) *Li, Ca, K* (4) *Ca, Li, K*

SPACE FOR ROUGH WORK

- | | |
|---|---|
| <p>68. KF combines with HF to form KHF_2. The compound contains the species
 (1) K^+, F^- and H^+ (2) K^+, F^- and HF
 (3) K^+ and $[\text{HF}_2]^-$ (4) $[\text{KHF}]^+$ and F^-</p> <p>69. The property of hydrogen which distinguishes it from other alkali metals is
 (1) Its electropositive character
 (2) Its affinity for non-metals
 (3) Its reducing character
 (4) Its non-metallic character</p> <p>70. Which of the following is strongest reducing agent
 (1) Li (2) K (3) Na (4) Rb</p> <p>71. The commercial production of sodium carbonate is done by
 (1) Lead-chamber process
 (2) Haber's process
 (3) Solvay's process
 (4) Castner's process</p> <p>72. On dissolving moderate amount of sodium metal in liquid at low temperature, which one of the following does not occur
 (1) Blue coloured solution is obtained
 (2) Na^+ ions are formed in the solution
 (3) Liquid becomes good conductor of electricity
 (4) Liquid ammonia remains diamagnetic</p> <p>73. Gobar salt is
 (1) $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$
 (2) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
 (3) $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$
 (4) $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$</p> <p>74. The colour given to the flame by sodium salts is
 (1) Light red (2) Golden yellow
 (3) Green (4) Pink</p> | <p>75. Photoelectric effect is maximum in
 (1) Cs (2) Na (3) K (4) Li</p> <p>76. Which of the following pair can't exist in solution
 (1) NaHCO_3 and NaOH
 (2) Na_2CO_3 and NaOH
 (3) Na_2CO_3 and NaCl
 (4) NaHCO_3 and NaCl</p> <p>77. The useful by-products, obtained in the Solvay process of manufacturing sodium carbonate, are
 (1) NaCl
 (2) NaHCO_3 and NH_4Cl
 (3) NH_4Cl Solution and quick lime
 (4) NaHCO_3 and CO_2</p> <p>78. The sum of protons, electrons and neutrons in the heaviest isotope of hydrogen is
 (1) 6 (2) 5
 (3) 4 (4) 3</p> <p>79. The colour of hydrogen is
 (1) Black (2) Yellow
 (3) Orange (4) Colourless</p> <p>80. Hydrogen from HCl can be prepared by
 (1) Mg (2) Cu
 (3) P (4) Pt.</p> <p>81. The difference between heat of adsorption of ortho and para hydrogen is
 (1) 0.4 kJ mol^{-1} (2) 0.8 kJ mol^{-1}
 (3) Zero (4) None of these</p> <p>82. Systematic name of H_2O (oxide of hydrogen) is
 (1) Water (2) Hydrogen oxide
 (3) Oxidane (4) None of these</p> |
|---|---|

SPACE FOR ROUGH WORK

83. Heavy water (D_2O) is
- (1) A product of oxygen and hydrogen
 - (2) Water of mineral springs
 - (3) Water obtained by repeated distillation and condensation
 - (4) Ordinary water containing dissolved salts heavy metals
84. Temporary hardness may be removed from water by adding
- (1) $CaCO_3$ (2) $Ca(OH)_2$
 - (3) $CaSO_4$ (4) HCl
85. D_2O is used more in
- (1) Chemical industry
 - (2) Nuclear reactor
 - (3) Pharmaceutical preparations
 - (4) Insecticide preparation
86. Which of the following is not true
- (1) Ordinary water is electrolysed more rapidly than D_2O
 - (2) Reaction between H_2 and Cl_2 is much faster than D_2 and Cl_2
 - (3) D_2O freezes at lower temperature than H_2O
 - (4) Bond dissociation energy for D_2 is greater than H_2
87. Match list I with list II and select the correct answer using the codes given below the lists

	List I	List II
1	Heavy water	(a) Bicarbonates of Mg and Ca in water
2	Temporary hard water	(b) No foreign ions in water
3	Soft water	(c) D_2O
4	Permanent hard water	(d) Sulphates and chlorides of Mg and Ca in water

Codes

- (1) 1-c, 2-d, 3-b, 4-a
 - (2) 1-b, 2-a, 3-c, 4-d
 - (3) 1-b, 2-d, 3-c, 4-a
 - (4) 1-c, 2-a, 3-b, 4-d
88. In which of the following reaction hydrogen peroxide is a reducing agent
- (1) $2FeCl_2 + 2HCl + H_2O_2 \rightarrow 2FeCl_3 + 2H_2O$
 - (2) $Cl_2 + H_2O_2 \rightarrow 2HCl + O_2$
 - (3) $2HI + H_2O_2 \rightarrow 2H_2O + I_2$
 - (4) $H_2SO_3 + H_2O_2 \rightarrow H_2SO_4 + H_2O$
89. H_2O_2 is always stored in black bottles because
- (1) It is highly unstable
 - (2) Its enthalpy of decomposition is high
 - (3) It undergo autooxidation on prolonged standing
 - (4) None of these
90. Correct solubility order is
- (1) $Na_2SO_4 > K_2SO_4 > Rb_2SO_4 > Cs_2SO_4$
 - (2) $K_2SO_4 < Na_2SO_4 < Rb_2SO_4 < Cs_2SO_4$
 - (3) $Na_2SO_4 < K_2SO_4 < Rb_2SO_4 < Cs_2SO_4$
 - (4) $Cs_2SO_4 < Na_2SO_4 < Rb_2SO_4 < K_2SO_4$

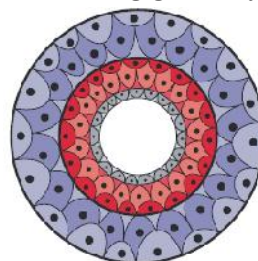
SPACE FOR ROUGH WORK

91. Which of the following statement / character is incorrect about the diagram given above?



- (1) Posses lateral appendages, parapodia which help in reproduction
(2) Body surface distinctly marked out into segments
(3) Circulatory system is closed
(4) Nephridia help in excretion and osmoregulation
92. In the evolution of animals a heart to pump the blood is found for the first time in
(1) Annelids (2) Round worms
(3) Arthropods (4) Flat worms
93. The heart of cockroach is
(1) Myogenic (2) Neurogenic
(3) Neuro-myogenic (4) None of the above
94. File like rasping organ in buccal cavity of molluscan is :
(1) Mantle (2) Radula
(3) Ctenidia (4) Pallium
95. Hemichordates do not have ?
(1) Stomochord
(2) Notochord
(3) Proboscis gland for excretion
(4) Buccal diverticulum
96. Chordates are different from non chordates that chordates have ?
(1) Dorsal, hollow, single nerve cord
(2) Pharynx perforated by gill slits
(3) Notochord
(4) All of these
97. Retrogressive metamorphosis is the character of:
(1) Cephalochordates (2) Hemichordates
(3) Urochordates (4) Cyclostomates
98. What is common to Earthworm, nereis and Leech:-
(1) Nephridia for excretion
(2) Metamerism
(3) True Coelomate
(4) All of above
99. Larva of Lamprey is
(1) Ammocoete (2) Trochophore
(3) Absent (4) Bipinnaria
100. The excretory structure of Echniodermata are:-
(1) Malpighian tubules (2) Green glands
(3) Kabers organ (4) Absent
101. In which one of the following all four sets of animals give birth to young ones ?
(1) Kangaroo, Dolphin, Scorpion
(2) Lion, Whale, Ostrich, Bat
(3) Penguin, Bat, *Hippopotamus*, Platypus
(4) Shrew, Bat, Cat, Frog
102. Count the correct number of statements which are not related to the members of class chondrichthyes
(a) terminal mouth
(b) placoid scales on skin
(c) presence of air bladder
(d) poikilothermal
(e) internal fertilization
(f) vivipary
(1) one (2) two (3) three (4) Four
103. Which of the following is not a trait of marsupials ?
(1) Oviparous
(2) Presence of teats over mammary gland
(3) Presence of brood pouch
(4) All of the above
104. Osteichthyes can be differentiated from chondrichthyes externally by the presence of
(1) Ventral mouth
(2) Lateral line system
(3) Terminal mouth
(4) Clasper
105. Which of the following is a true Amphibian ?
(1) *Testudo* (2) *Crocodylus*
(3) *Hyla* (4) *Scoliodon*

106. Which of the following is not an example of fish ?
 (1) *Scoliodon* (2) *Carcharodon*
 (3) *Ichthyophis* (4) *Clarius*
107. The most unique mammalian characteristic is the
 (1) Mammary glands
 (2) Internal fertilisation
 (3) Warm blooded
 (4) Heart is four chambered
108. Pearl is made up of-
 (1) CaCO_3 (2) Na_2CO_3
 (3) MgCO_3 (4) NaCl
109. A true coelom (Enterocoel) is absent in phylum
 (1) Annelida (2) Echinodermata
 (3) Hemichorata (4) Chordata
110. Cell aggregate type body plan found in :-
 (1) Cnidaria (2) Nematoda
 (3) Porifera (4) Arthropoda
111. Deuterostomes are
 (1) Echinodermata (2) Arthropoda
 (3) Annelida (4) Platyhelminthes
112. An opening serving both as mouth and anus is found in
 (1) Annelids and Molluscs
 (2) Coelenterates and Annelids
 (3) Coelenterates and Platyhelminthes
 (4) Coelenterates and Aschelminthes
113. True coelom is lined by
 (1) Ectoderm (2) Endoderm
 (3) Mesoglea (4) Mesoderm
114. A coelom is absent in
 (1) Coelenterata (2) Porifera
 (3) Platyhelminthes (4) All the above
115. If Female reproductive part mature early, in bisexual animal the phenomenon is called
 (1) Sexual dimorphism (2) Protandry
 (3) Metagenesis (4) Protogyny
116. The symmetry found in an animal which can be divided into two similar halves by one plane only, into right and left mirror images is
 (1) Biradial (2) Spherical
 (3) Radial (4) Bilateral
117. Which of the following is the only triploblastic acoelomate phylum?
 (1) Porifera (2) Cnidaria
 (3) Ctenophora (4) Platyhelminthes
118. Symmetry of Ctenophores is generally
 (1) Spherical (2) Bilateral
 (3) Biradial (4) Asymmetrical
119. A coelom formed by fusion of gut pouches called
 (1) Schizocoel (2) Haemocoel
 (3) Enterocoel (4) Pseudocoel
120. Cnidarians and Acnidarians share common characters like ?
 (1) Diploblastic
 (2) Both extracellular & intracellular digestion
 (3) Tissue level of organisation
 (4) All of the above
121. Notochord is a rod like structure found on the dorsal side during embryonic development in some animals and is derived from
 (1) Mesoderm (2) Endoderm
 (3) Ectoderm (4) Both 2 & 3
122. In which of the following bilateria, the body cavity is not lined by mesoderm. Instead, the mesoderm is present as scattered pouches in between the ectoderm and endoderm?
 (1) Echinoderms (2) Annelids
 (3) Platyhelminthes (4) Aschelminthes
123. In which organisms, the body can be divided into identical left and right halves in only one plane in only larval stage?
 (1) Arthropoda (2) Mollusca
 (3) Annelida (4) Echinodermata
124. Metameric segmentation is present in
 (1) Annelids and Mollusca
 (2) Annelids and Arthropoda
 (3) Porifera and Coelenterata
 (4) Platyhelminthes and Nematoda
125. Examine the diagrammatic sectional view of an animals showing germ layers and coelom.



Choose the correct explanation about it
 (1) Triploblastic and acoelomate
 (2) Triploblastic and coelomate
 (3) Diploblastic and acoelomate
 (4) Diploblastic animal having the thickest layer as mesoglea

126. Match the following list of animals with their level of organization.

Level of organisation Animal

- | | |
|------------------------|----------------------|
| (A) Organ level | (i) Annelida |
| (B) Cellular level | (ii) Platyhelminthes |
| (C) Tissue level | (iii) Sponge |
| (D) Organ system level | (iv) Cnidarian |

Choose the correct match.

- (1) A-(iv), B-(i), C-(ii), D-(iii)
 (2) A-(iv), B-(i), C-(iii), D-(ii)
 (3) A-(ii), B-(iii), C-(iv), D-(i)
 (4) A-(i), B-(iv), C-(iii), D-(ii)

127. Which arrangement is in correct descending order :-

- (1) Kingdom > Phylum > Order > Class
 (2) Phylum > Class > Order > Kingdom
 (3) Order > Class > Kingdom > Phylum
 (4) Kingdom > Phylum > Class > Order

128. Exoskeleton of CaCO_3 is found in which protozoan :-

- (1) Amoeba (2) Foraminiferans
 (3) Radiolarians (4) Paramecium

129. Which of the following is characteristic feature of mastigophora?

- (1) Cilia (2) Food vacuole
 (3) Flagella (4) Pellicle

130. Match the column-I with column-II and select the correct answer :-

Column-I

Column-II

- | | |
|-------------------------|--------------------|
| A. Giardia intestinalis | i. Pebrine disease |
| B. Leishmania tropica | ii. Pyorrhoea |
| C. Nosema | iii. Giardiasis |
| D. Trichomonas tinax | iv. Oriental sore |

- (1) A-iii, B-ii, C-i, D-iv
 (2) A-ii, B-iv, C-i, D-iii
 (3) A-iii, B-iv, C-i, D-ii
 (4) A-i, B-iv, C-iii, D-ii

131. Macronucleus of Paramecium is responsible for :-

- (1) Metabolic activities
 (2) Genetical activities
 (3) Sexual Reproduction
 (4) Both (2) and (3)

132. Which organism is considered as bell animacule?

- (1) Amoeba (2) Euglena
 (3) Trypanosoma (4) Vorticella

133. In which of the animal dimorphic nucleus is found?

- (1) Amoeba (2) Trypanosoma
 (3) Plasmodium (4) Paramecium

134. African sleeping sickness by

- (1) Tse Tse fly (2) Dragon fly
 (3) Sand fly (4) Fruit fly

135. Germ cells of sponges are formed by

- (1) Choanocytes (2) Collenocytes
 (3) Chromocytes (4) Archaeocytes

136. Common bath sponge is

- (1) Sycon (2) Euspongia
 (3) Spongilla (4) Cliona

137. Sponges capture food by

- (1) Trophocytes (2) Pinacocytes
 (3) Choanocytes (4) Silicoblasts

138. In porifera, skeleton forming cells are

- (1) Thesocytes (2) Myocytes
 (3) Archaeocytes (4) Scleroblast

139. Water current in a sponge is maintained by

- (1) Amoebocytes (2) Myocytes
 (3) Pinacocytes (4) Choanocytes

140. Choanocytes are

- (1) Biflagellated (2) Multiflagellated
 (3) Uniflagellated (4) Without flagella

141. Smallest sponge is

- (1) Leucosolenia (2) Chalina
 (3) Cliona (4) Poterion

142. Study of sponges is known as

- (1) Protozoology (2) Parazoology
 (3) Helminthology (4) Entomology

143. Sponges are

- (1) Mostly marine (2) Hermaphrodite
 (3) Mostly asymmetrical
 (4) All of these

144. Pinacoderm layer of body wall in sponges has

- (1) Pinacocytes (2) Porocytes
 (3) Both (1) and (2) (4) Choanocytes

145. In a sponge, digestion takes place in

- (1) Osculum (2) Ostium
 (3) Paragastric cavity
 (4) Food vacuoles of Choanocytes

146. Sponges have great regenerating power due to which are cells

- (1) Archaeocytes, flagellated
 (2) Archaeocytes, totipotent
 (3) Scleroblasts, amoeboid
 (4) Choanocytes, flagellated

147. Which of the following cannot be considered as function of canal system of sponges?
- Respiratory exchange
 - Formation of gametes
 - Removal of waste
 - Food gathering
148. Match the following.
- | Column I | Column II |
|------------------|----------------------------|
| (A) Archaeocytes | (i) For food storage |
| (B) Thesocytes | (ii) For food distribution |
| (C) Scleroblast | (iii) Totipotent cells |
| (D) Trophocytes | (iv) Secrete Skeleton |
- A-(iii), B-(i), C-(iv), D-(ii)
 - A-(i), B-(ii), C-(iii), D-(iv)
 - A-(i), B-(iii), C-(ii), D-(iv)
 - A-(iii), B-(iv), C-(i), D-(ii)
149. Larva of Hydra is
- Planula
 - Hydrula
 - Absent
 - Ephyra
150. Hydra is
- Diploblastic, Bilaterally symmetrical and acoelomate
 - Diploblastic, Radially symmetrical and acoelomate
 - Triploblastic, Bilaterally symmetrical and acoelomate
 - Triploblastic, Radially symmetrical and acoelomate
151. Digestion in Hydra is
- Saprophytic
 - Extra and intracellular
 - Intracellular only
 - Extracellular only
152. Mark the cell which first time evolved in Coelenterata
- Nerve cell
 - Muscle cell
 - Gland cell
 - Germ cell
153. The number of testicular lobules in each testis are
- 400
 - 500
 - 20
 - 250
154. Sperms are produced in
- Seminal vesicles
 - Prostate gland
 - Seminiferous tubules
 - Epididymis
155. Testis are suspended in a pouch of skin called
- Inguinal canal
 - Scrotum
 - Tunica vasculosa
 - Bladder
156. Seminiferous tubules are lined by two types of cells, i.e.
- Sertoli cells and Leydig cells
 - Sertoli cells and Nurse cells
 - Spermatogonial cells and Sertoli cells
 - Spermatogonial cells and Leydig cells
157. The Leydig cells secrete
- Progesterone
 - Estrogen
 - Testosterone
 - Cortisol
158. Sugars that sperm use for energy is secreted by
- Prostate gland
 - Seminal vesicle
 - Vasa efferentia
 - Epididymis
159. Testis is divided into lobules due to partitions that have developed from
- Tunica media
 - Tunica vasculosa
 - Tunica albuginea
 - Tunica interna
160. The outermost covering of the testis is
- Gubernaculum
 - Tunica vasculosa
 - Tunica vaginalis
 - Tunica albuginea
161. The percentage of semen contributed by seminal vesicles is
- 30%
 - 50%
 - 70%
 - 90%
162. The unpaired gland in male reproductive system is
- Cowper's gland
 - Seminal vesicle
 - Bulbourethral gland
 - Prostate gland
163. Which of the following is the correct path of movement of sperms?
- Seminiferous tubule → vasa efferentia → rete testis → caput epididymis → tubuli recti
 - Seminiferous tubule → rete testis → vasa efferentia → tubuli recti → cauda epididymis
 - Seminiferous tubule → tubuli recti → rete testis → vasa efferentia → caput epididymis → corpus epididymis
 - Seminiferous tubule → caput epididymis → vasa efferentia → corpus epididymis → rete testis

- 164.** Which of the following function of Sertoli cell mediates a negative feedback control of Follicle Stimulating Hormone (FSH) secretions?
- (1) Formation of testosterone
 - (2) Formation of anti-mullerian factor
 - (3) Formation of blood testis barrier
 - (4) Synthesis of inhibin
- 165.** Birth canal comprises of
- (1) Vaginal canal only
 - (2) Cervical canal only
 - (3) Fallopian tube
 - (4) Vaginal canal and cervical canal
- 166.** The space between two folds of labia minora is called
- (1) Perineum
 - (2) Vestibule
 - (3) Clitoris
 - (4) Foreskin
- 167.** In females, an erectile body homologous with penis of male is
- (1) Prepuce
 - (2) Fourchette
 - (3) Clitoris
 - (3) Uterus
- 168.** Which of the following is not a male secondary sexual character?
- (1) Enlarged penis
 - (2) Hoarse voice
 - (3) Beard and moustache
 - (4) Deposition of fats in buttocks
- 169.** Mammary glands are
- (1) Modified sebaceous gland
 - (2) Modified sweat gland
 - (3) Homologous to male prostate
 - (4) Endocrinal glands
- 170.** Deposition of fats in the breast, hips as well as growth of pubic hair during puberty is under the influence of
- (1) Androgen
 - (2) Progesterone
 - (3) Estrogen
 - (4) LH
- 171.** In mammary glands milk is sucked out through
- (1) Ampulla
 - (2) Alveoli
 - (3) Lactiferous duct
 - (4) Mammary ductules
- 172.** Which of the following structure is/are present in vestibule part of female reproductive system?
- (i) Clitoris
 - (ii) Perineum
 - (iii) Urethral orifice
 - (iv) Vaginal orifice
- (1) (i) and (ii)
 - (2) (iii) and (iv)
 - (3) (ii) and (iii)
 - (4) (i) and (iv)
- 173.** The cervical canal opens into vagina by an opening called
- (1) Internal Os
 - (2) External Os
 - (3) Ampulla
 - (4) Fundus
- 174.** Absence of menstrual cycle is indicative of
- (1) Pregnancy
 - (2) Poor health
 - (3) Stress
 - (4) All of above
- 175.** Luteal phase of menstrual cycle is also known as
- (1) Proliferative phase
 - (2) Secretory phase
 - (3) Follicular phase
 - (4) Ovulatory phase
- 176.** Estrous cycle is characteristic of
- (1) Human females
 - (2) All vertebrates
 - (3) All mammals
 - (4) Non-primate mammalian females
- 177.** Which of the following is also called 'yellow body'?
- (1) Corpus albicans
 - (2) Corpus luteum
 - (3) Graafian follicle
 - (4) Cumulus oophoricus
- 178.** If the menstrual cycle of a woman is of 39 days, then ovulation occurs on
- (1) 25th day
 - (2) 14th day
 - (3) 20th day
 - (4) 7th day
- 179.** During ovulation in human female
- (1) Three peaks of oestrogen is present
 - (2) Peak of FSH is higher than LH
 - (3) Peak of LH is higher than FSH
 - (4) Tallest peak of FSH is observed
- 180.** Antrum, a fluid-filled cavity is typically seen in which follicle?
- (1) Primordial follicle
 - (2) Primary follicle
 - (3) Secondary follicle
 - (4) Graafian follicle